📐 EduTrack — System Design Document

Version 1.0 | VTU Internship 2026 — Team 15 | April 2026

1. Overview

EduTrack is a full-stack academic management platform connecting students, teachers, and administrators. It provides real-time management of attendance, assignments, marks, course materials, and system settings through a role-based web application.

2. System Architecture

2.1 High-Level Architecture

```
┌──────────────────────────────────────────────────────────────┐
│                          CLIENT                              │
│                React 19 (Vercel CDN)                         │
│   https://vtu-intern-2026-team15-java-fs.vercel.app          │
└──────────────────────────┬───────────────────────────────────┘
                           │ HTTPS REST API calls
┌──────────────────────────▼───────────────────────────────────┐
│                         BACKEND                              │
│            Spring Boot 3 (Render - Docker)                   │
│        https://edutrack-backend-o1ym.onrender.com            │
└──────────────────────────┬───────────────────────────────────┘
                           │ JDBC (MySQL Connector)
┌──────────────────────────▼───────────────────────────────────┐
│                        DATABASE                              │
│               MySQL 9.4 (Railway Cloud)                      │
│            hopper.proxy.rlwy.net:12491                       │
└──────────────────────────────────────────────────────────────┘
```

2.2 Deployment

LAYER	TECHNOLOGY	PLATFORM
Frontend	React 19	Vercel
Backend	Spring Boot 3 / Docker	Render
Database	MySQL 9.4	Railway

3. Technology Stack

Frontend

COMPONENT	TECHNOLOGY
Framework	React 19
Routing	Hash-based (window.history)
HTTP Client	Fetch API
Testing	Playwright E2E — 43 tests
Build	Create React App

Backend

COMPONENT	TECHNOLOGY
Framework	Spring Boot 3.2.3
Language	Java 21
ORM	Spring Data JPA / Hibernate
Security	Spring Security + BCrypt
Email	Spring Mail (Gmail SMTP)
Build	Maven + Docker
Testing	JUnit 5 + Mockito — 31 tests

Database

COMPONENT	TECHNOLOGY
RDBMS	MySQL 9.4

COMPONENT	TECHNOLOGY
Hosting	Railway Cloud

4. Database Design

Tables

admins

COLUMN	TYPE	NOTES
admin_id	INT PK	Auto increment
name	VARCHAR(100)	Not null
email	VARCHAR(100)	Unique
password	VARCHAR(255)	BCrypt hashed

teachers

COLUMN	TYPE	NOTES
teacher_id	INT PK	Auto increment
name	VARCHAR(100)	Not null
email	VARCHAR(100)	Unique
password	VARCHAR(255)	BCrypt hashed
phone	VARCHAR(15)	
department	VARCHAR(100)	Not null
employee_id	VARCHAR(50)	Unique
status	VARCHAR(20)	Default: Active

students

COLUMN	TYPE	NOTES
student_id	INT PK	Auto increment
name	VARCHAR(100)	Not null

COLUMN	TYPE	NOTES
email	VARCHAR(100)	Unique
password	VARCHAR(255)	BCrypt hashed
phone	VARCHAR(15)	
usn	VARCHAR(50)	Unique
department	VARCHAR(100)	Not null
year	VARCHAR(20)	Not null
semester	VARCHAR(20)	
status	VARCHAR(20)	Default: Active
course_id	INT FK	→ courses

courses

COLUMN	TYPE	NOTES
course_id	INT PK	Auto increment
course_name	VARCHAR(150)	Not null
credits	INT	Default: 3
teacher_id	INT FK	→ teachers (CASCADE)

attendance

COLUMN	TYPE	NOTES
attendance_id	INT PK	Auto increment
student_id	INT FK	→ students (CASCADE)
course_id	INT FK	→ courses (CASCADE)
attendance_date	DATE	Not null
status	VARCHAR(10)	Present / Absent

assignments

COLUMN	TYPE	NOTES
assignment_id	INT PK	Auto increment
course_id	INT FK	→ courses (CASCADE)
title	VARCHAR(200)	Not null
description	TEXT	
due_date	DATE	Not null

submissions

COLUMN	TYPE	NOTES
submission_id	INT PK	Auto increment
assignment_id	INT FK	→ assignments (CASCADE)
student_id	INT FK	→ students (CASCADE)
submitted_at	DATETIME	Not null
file_url	VARCHAR(500)	
status	VARCHAR(20)	Default: Submitted
grade	VARCHAR(10)	
feedback	TEXT	

marks

COLUMN	TYPE	NOTES
mark_id	INT PK	Auto increment
student_id	INT FK	→ students (CASCADE)
course_id	INT FK	→ courses (CASCADE)
marks	INT	Default: 0
grade	VARCHAR(5)	Auto-calculated

course_materials

COLUMN	TYPE	NOTES
material_id	INT PK	Auto increment
course_id	INT FK	→ courses (CASCADE)
title	VARCHAR(200)	Not null
material_url	VARCHAR(500)	Not null
uploaded_at	DATETIME	Not null

otp_verification

COLUMN	TYPE	NOTES
id	BIGINT PK	Auto increment
email	VARCHAR(100)	Not null
otp	VARCHAR(6)	6-digit code
expiry_time	DATETIME	10 min from creation
used	BOOLEAN	Default: false

system_settings

COLUMN	TYPE	NOTES
id	INT PK	Fixed: 1
active_year	VARCHAR(20)	e.g. 2025-26
active_semester	VARCHAR(20)	e.g. Even
maintenance_mode	BOOLEAN	Default: false
registration_open	BOOLEAN	Default: true

5. API Design

Authentication

METHOD	ENDPOINT	DESCRIPTION
POST	/api/auth/login/admin	Admin login

METHOD	ENDPOINT	DESCRIPTION
POST	/api/auth/login/student	Student login
POST	/api/auth/login/teacher	Teacher login
POST	/api/auth/register/student	Student registration
POST	/api/auth/admin/add-student	Admin adds student
POST	/api/auth/admin/add-teacher	Admin adds teacher
POST	/api/auth/change-password/admin	Change admin password
POST	/api/auth/change-password/student	Change student password
POST	/api/auth/change-password/teacher	Change teacher password

OTP / Password Reset

METHOD	ENDPOINT	DESCRIPTION
POST	/api/otp/forgot-password	Send OTP (role-validated)
POST	/api/otp/verify	Verify OTP
POST	/api/auth/reset-password/student	Reset student password
POST	/api/auth/reset-password/teacher	Reset teacher password
POST	/api/auth/reset-password/admin	Reset admin password

Students

METHOD	ENDPOINT	DESCRIPTION
GET	/api/students	List all
GET	/api/students/{id}	Get by ID
PUT	/api/students/{id}	Update profile
DELETE	/api/students/{id}	Delete
PATCH	/api/students/{id}/status	Update status

Teachers

METHOD	ENDPOINT	DESCRIPTION
GET	/api/teachers	List all
GET	/api/teachers/{id}	Get by ID
PUT	/api/teachers/{id}	Update profile
DELETE	/api/teachers/{id}	Delete (cascades)
PATCH	/api/teachers/{id}/status	Update status

Courses

METHOD	ENDPOINT	DESCRIPTION
GET	/api/courses	List all
POST	/api/courses	Create
PUT	/api/courses/{id}	Update
DELETE	/api/courses/{id}	Delete (cascades)

Attendance

METHOD	ENDPOINT	DESCRIPTION
GET	/api/attendance/student/{id}	By student
GET	/api/attendance/course/{id}	By course
GET	/api/attendance/course/{id}/stats	Stats per student
POST	/api/attendance	Mark (upsert)

Marks

METHOD	ENDPOINT	DESCRIPTION
GET	/api/marks/student/{id}	By student
GET	/api/marks/course/{id}	By course
POST	/api/marks	Enter (auto-grade)

METHOD	ENDPOINT	DESCRIPTION
PUT	/api/marks/{id}	Update

System Settings

METHOD	ENDPOINT	DESCRIPTION
GET	/api/system/settings	Get settings
PUT	/api/system/settings	Update settings

6. Security Design

ASPECT	IMPLEMENTATION
Password storage	BCrypt (strength 10)
Authentication	Email + password match
OTP	6-digit, 10-min expiry, single-use
OTP validation	Role + email existence checked before sending
CSRF	Disabled (stateless REST)
CORS	All origins allowed for production
Authorization	Role-based frontend portals

7. Grading Logic

SCORE	GRADE
90–100	O
80–89	A+
70–79	A
60–69	B+
Below 60	B

8. Testing

LAYER	TOOL	COUNT
Backend Services	JUnit 5 + Mockito	31 tests
Frontend E2E	Playwright	43 tests
API	Postman Collection	26 requests

9. Live URLs

SERVICE	URL
Frontend	`https://vtu-intern-2026-team15-java-fs.vercel.app`
Backend	`https://edutrack-backend-o1ym.onrender.com`
GitHub	`https://github.com/NaveenPR123/VTU_INTERN_2026_Team15_JavaFS`